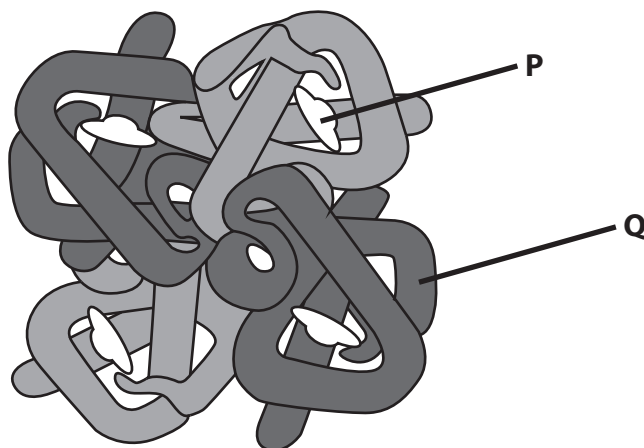


- 4 The role of haemoglobin is to transport oxygen and carbon dioxide in the blood.

The diagram shows the structure of adult haemoglobin



- (a) (i) State the role of the structure labelled **P**.

(1)

- (ii) Explain the properties of amino acids located on the outer surface of the haemoglobin, for example at position **Q**.

(2)

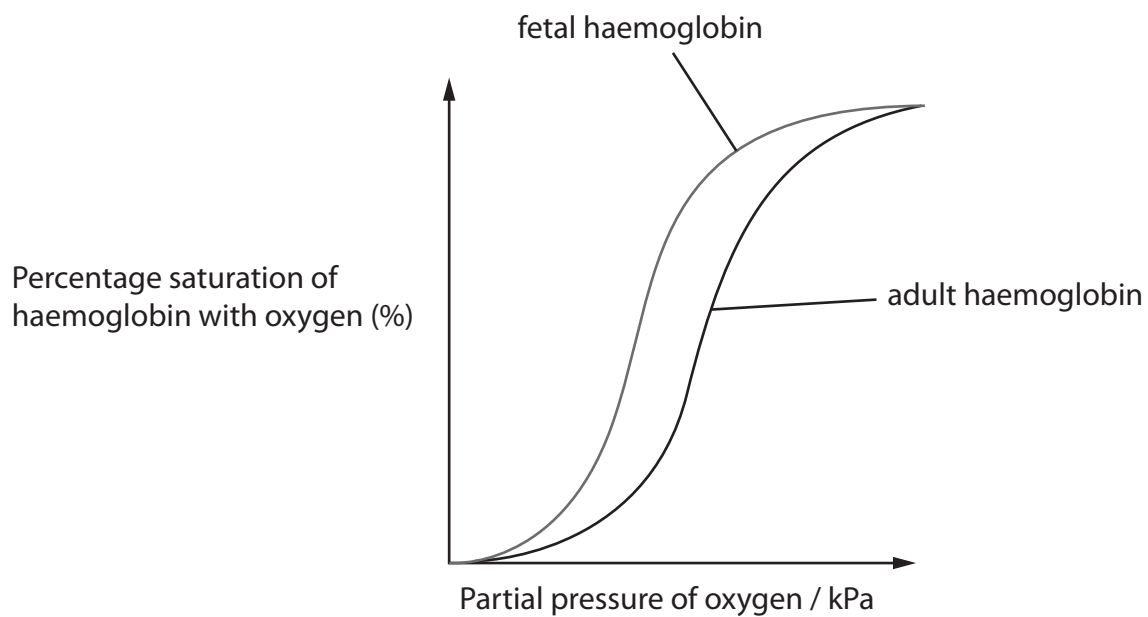
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(b) The graph shows the oxygen dissociation curves of adult and fetal haemoglobin.



- (i) Explain why the oxygen dissociation curve of adult haemoglobin is different from that of fetal haemoglobin.

(2)

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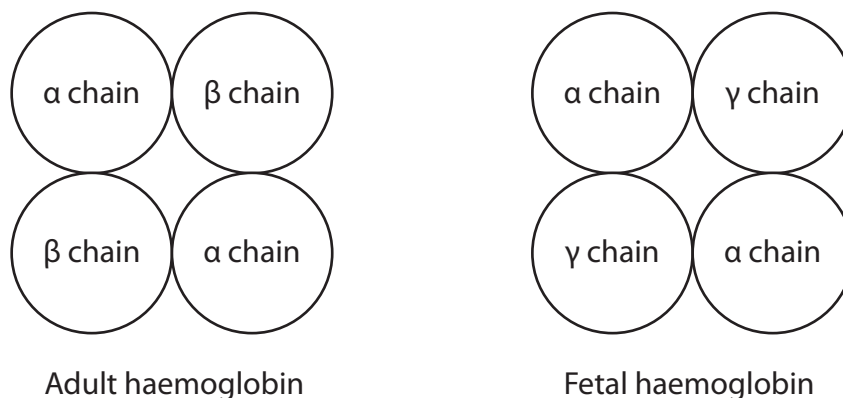
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(ii) The structures of adult and fetal haemoglobin are different.

The diagram shows the structure of adult and fetal haemoglobin.



The table shows the number of amino acids in each type of chain.

Type of chain	Number of amino acids
α	141
β	146
γ	146

The amino acids in the α chains are the same in adult and fetal haemoglobin. The β and γ chains differ in 39 of their amino acids.

Calculate the percentage of amino acids that are different in adult and fetal haemoglobin.

(2)

Answer %

(Total for Question 4 = 7 marks)

